



Smart City Traffic Management System Using AutoCAD and Simulation

Revolutionizing city mobility through intelligent simulation and real-time control.

Course: Engineering Drawing
Course Code: 07322110

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Introduction

- Rapid urban traffic growth
- Need for intelligent, automated systems
- AI + simulation enable proactive control
- Smart city traffic: safety, flow, resilience

Problem Statement



Split-screen: Traditional vs AI-driven control

- Traffic congestion — unrecoverable delays
- Delayed emergency response
- Manual signal timing limitations
- Poor situational monitoring → higher accidents

Project Objectives



Reduce Congestion

Adaptive signal timing and flow optimization



Improve Safety

Real-time incident detection and alerts



Intelligent Control

AI-driven signal decisions and routing



Emergency Routing

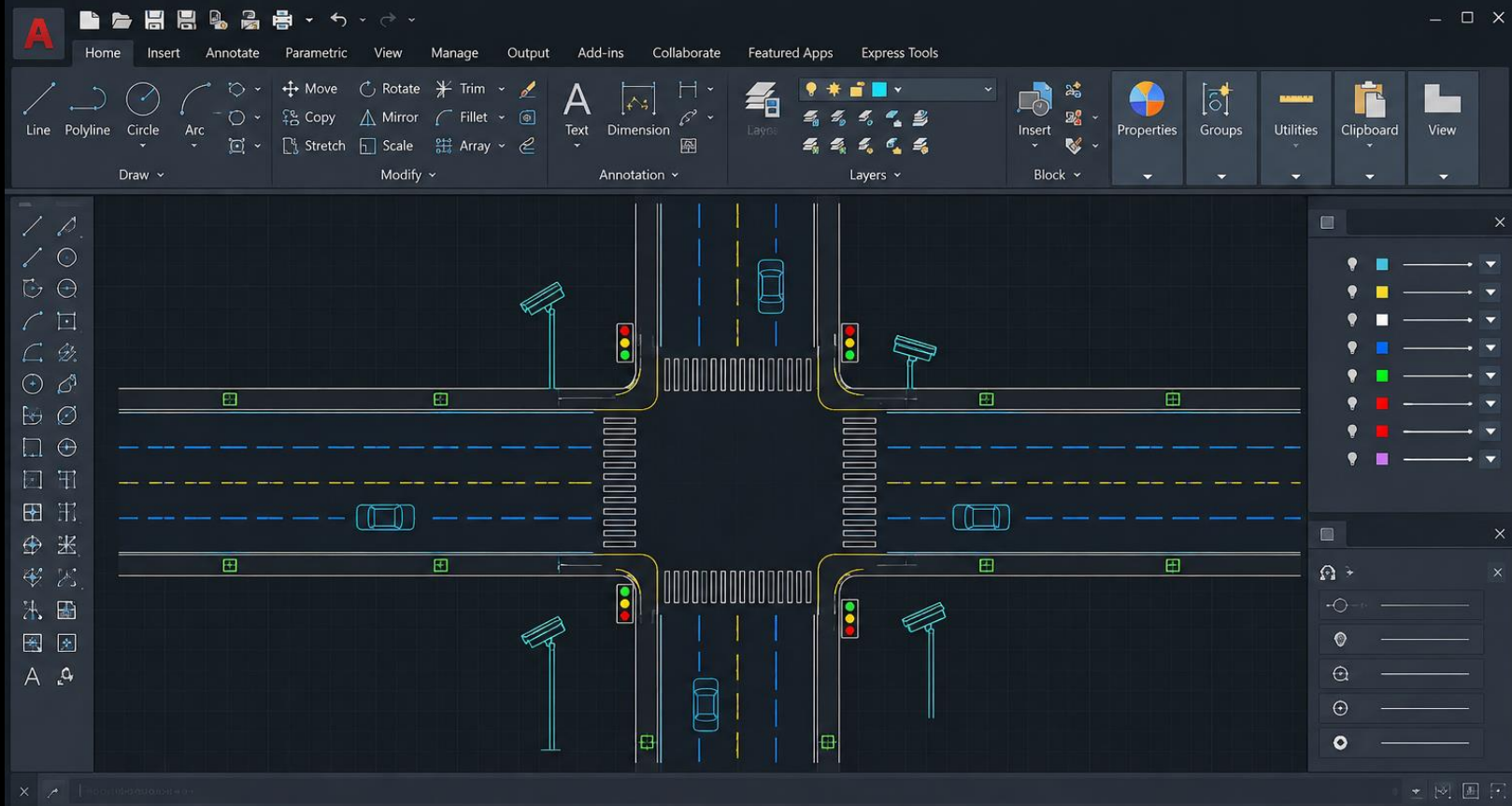
Priority corridors for first responders



Smart Monitoring

Drone/CCTV fusion for situational awareness

How AutoCAD is used in Our PROJECT



Engineering Drawing — AutoCAD

Road layout planning · Signal & CCTV
placement · Lane and junction design

Road Layout

Optimized lane
geometry,
reversible lanes

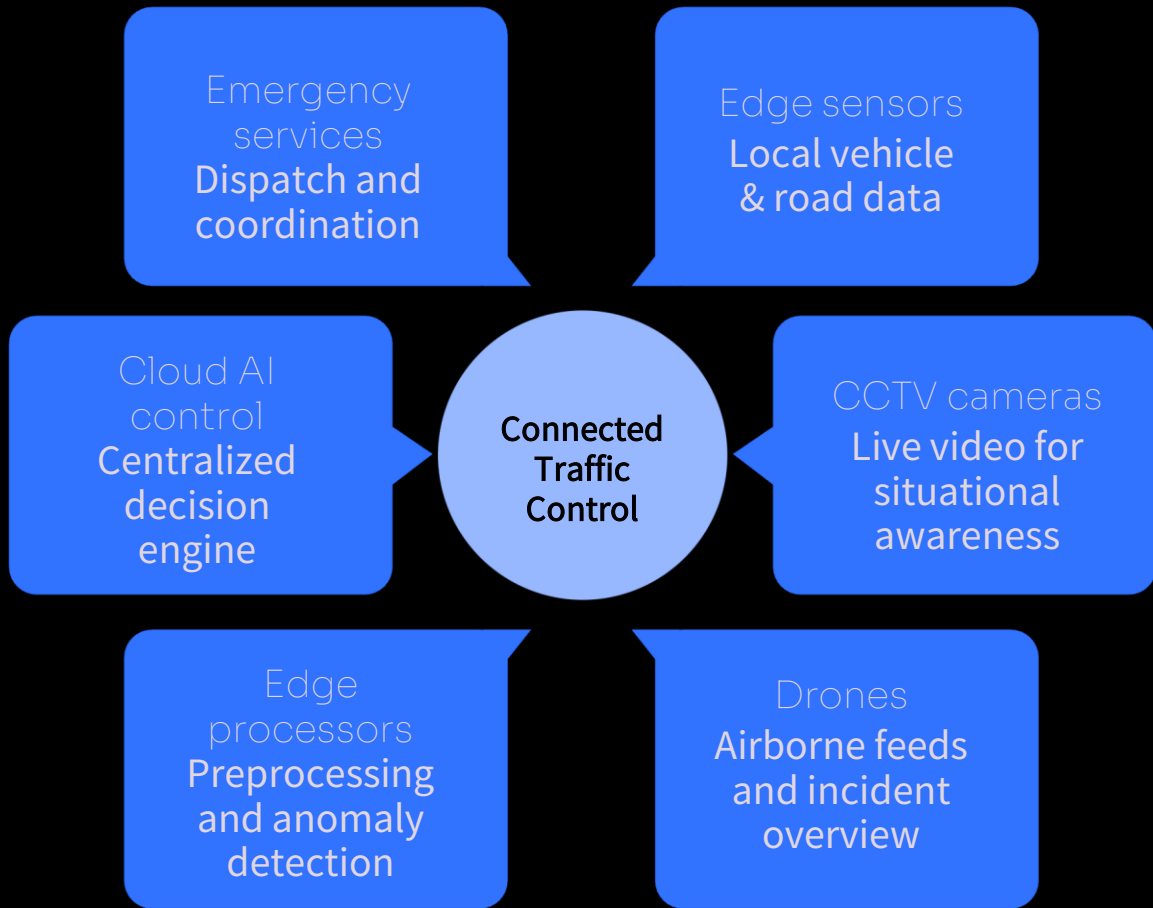
Signal
Positioning

Line-of-sight &
sensor coverage

CCTV &
Drones

Coverage zones &
overlapping
sensors





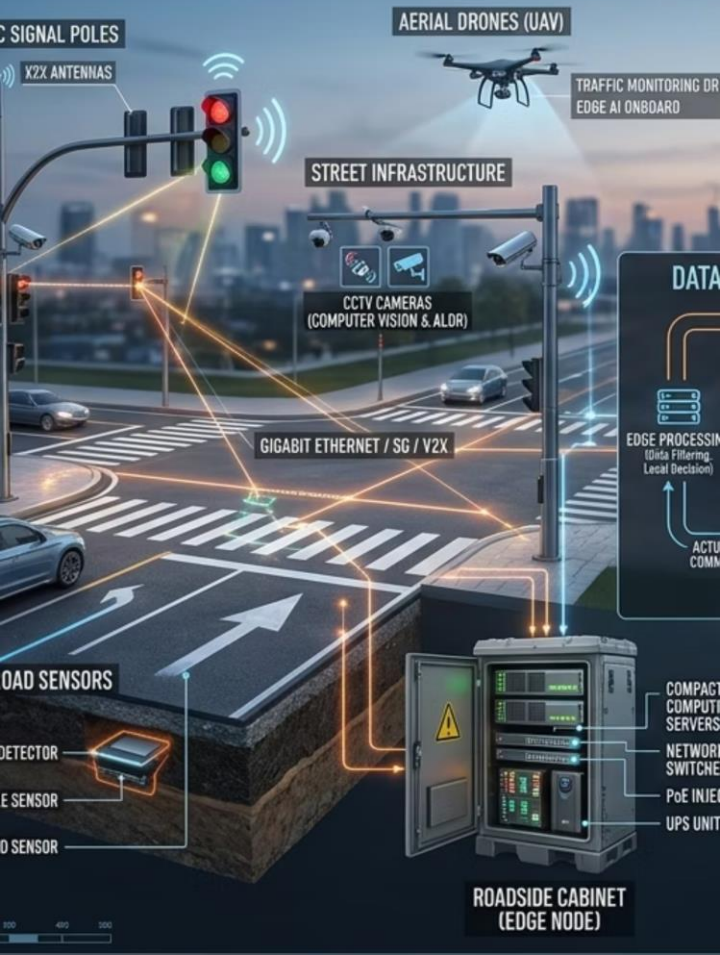
System Architecture

- Edge sensors & cameras → local preprocessing
- Secure data stream → Cloud AI controller
- AI issues signal decisions → Edge actuators
- Emergency services & drone feed integration

AI Traffic Simulation

- Vehicle detection & density heatmaps
- Adaptive signal timing via reinforcement learning
- Priority routing for emergency vehicles
- What-if scenarios: events, accidents, road works





Hardware & Software Components

Hardware

Smart signals •
CCTV • Drones •
IoT sensors • Edge
servers

Software

AutoCAD • Traffic
simulator • ML
models • Cloud
orchestration

Advantages & Future Improvements

Advantages

- Reduced congestion & wait times
- Improved safety & incident response
- Data-driven urban planning

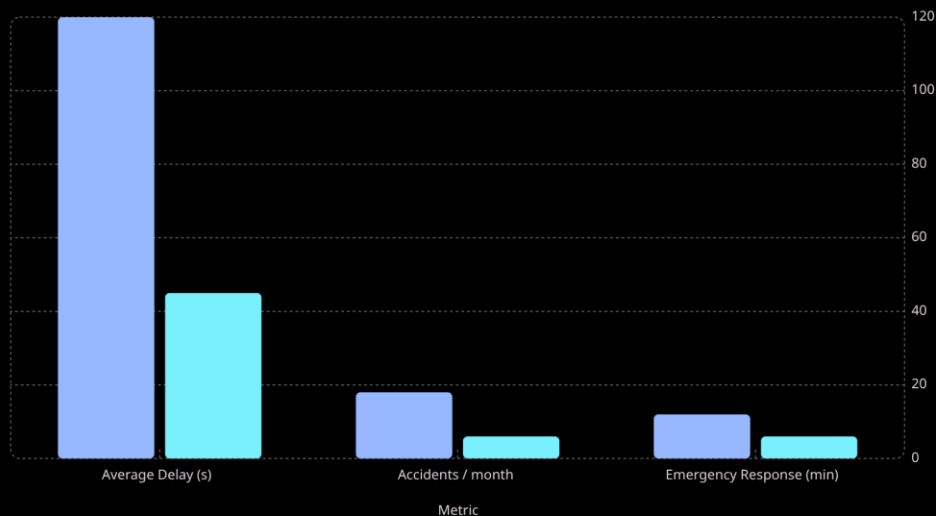
Future Improvements

- AI prediction models for demand
- Autonomous vehicle integration
- IoT-enabled smart roads & cloud monitoring

📄 Roadmap visual: short-term → mid-term → long-term



BEFORE AFTER



Results & Impact

- Efficient traffic management — measurable gains
- Reduced waiting time & fewer incidents
- Scalable planning inputs for city authorities

Visual: before vs after cinematic roadway comparison

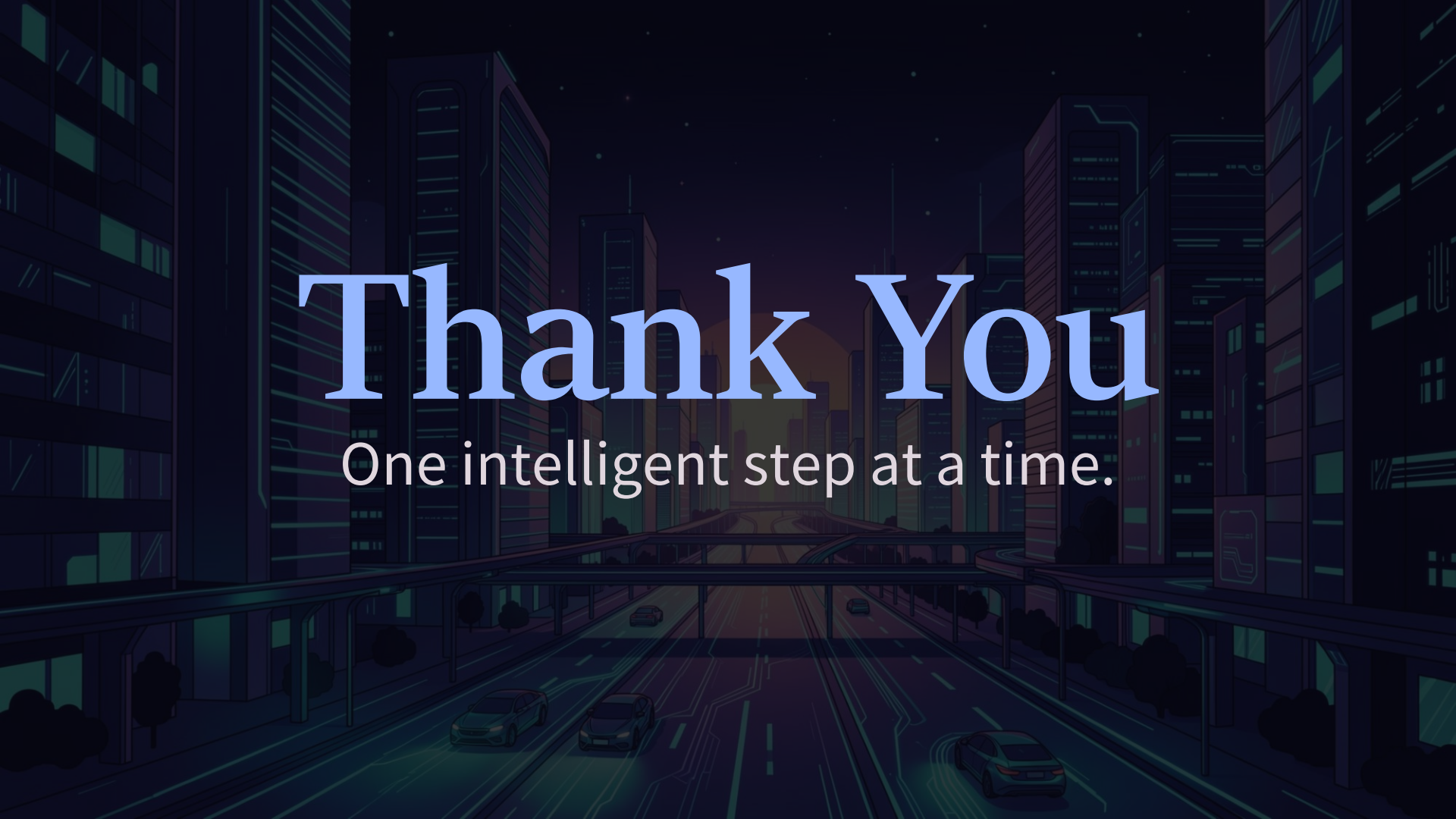


Future Vision of Smart Traffic Management

AI-Powered Traffic Control
Smart Roads, Smart Future.
Real-Time Traffic Intelligence
Safer & Faster Urban Mobility

A dark, futuristic city street at night, illuminated by blue and purple light trails. Two cars are visible on the road, one in the foreground and one further back. Above the cars, several drones are flying, and a large, glowing, circular digital interface is visible in the center of the sky. The background features tall, dark buildings with glowing windows and various digital screens and data visualizations floating in the air.

Questions & Answers

The background is a stylized illustration of a city street at night. Tall, dark buildings line both sides of the street, with some windows glowing. A multi-lane road with a central divider and overpasses stretches into the distance. Several cars are visible on the road, some with motion blur. The overall color palette is dark, with blues, purples, and greys, accented by the white text.

Thank You

One intelligent step at a time.